

Surgical Management of Breast Cancer

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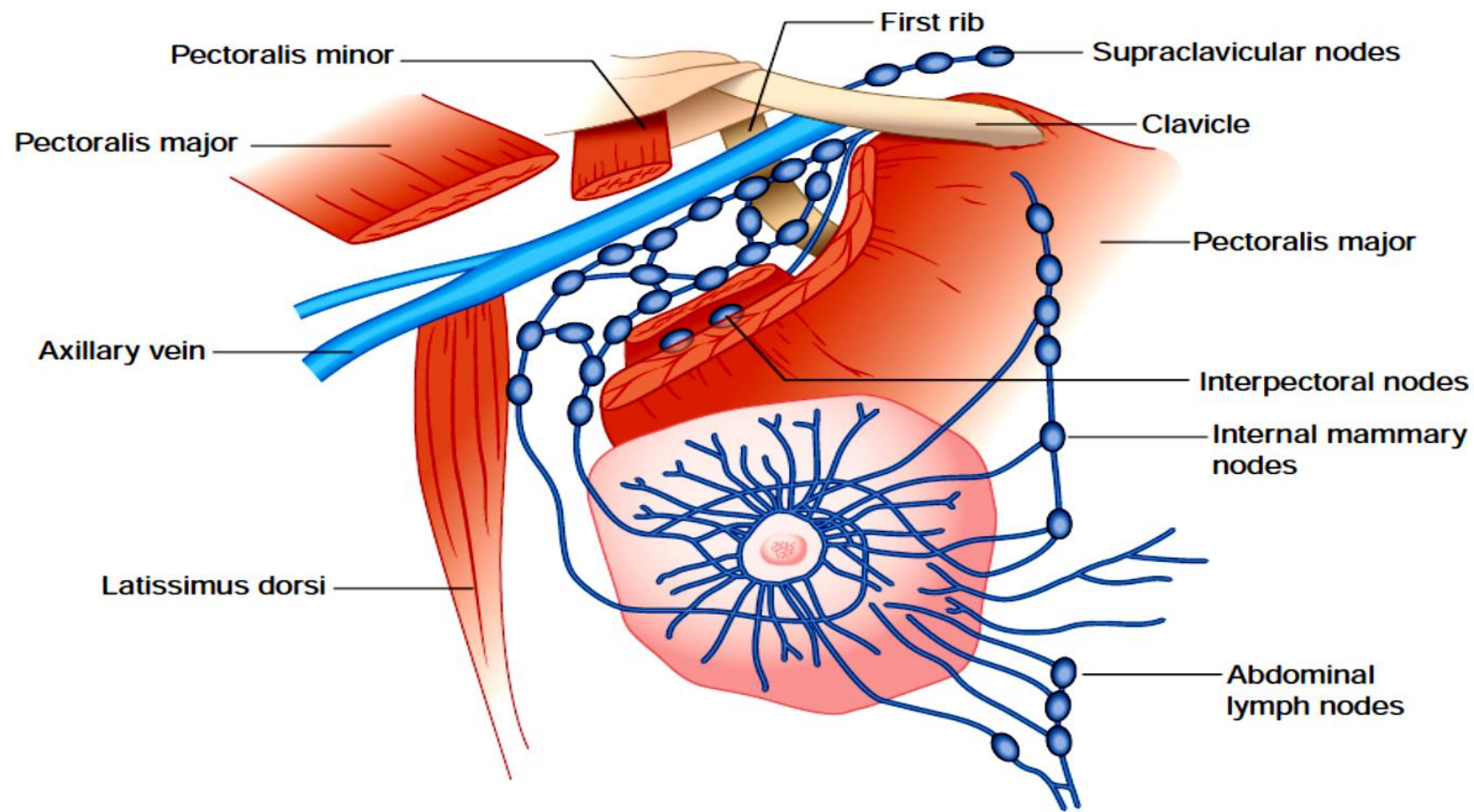


Key facts



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- Total of 35 000 new cases per year; 1 in 9 lifetime risk for women.
- Commonest in Western Europe; least common in Japan and Africa.
- Incidence increases with age.
- 1 % occurs in men.
- 5 % related to identifiable genetic abnormality (BRAC1,BRAC2, ataxia–telangectasia genes.)
- 60 % present as symptomatic disease; 40% during screening.



Lymph node drainage of the breast.

Pathological features

80 % ductal
adenocarcinoma

20% lobular,
mucinous tubular
or medullary
adenocarcinoma.

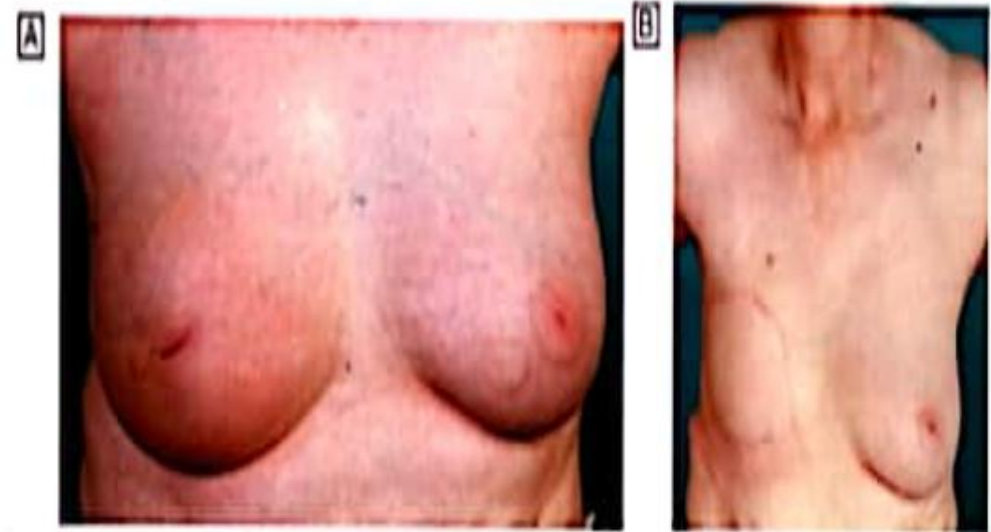
Most carcinomas
believed to originate
as *in situ* carcinoma
before becoming
invasive;

70% express
oestrogen or
progesterone
receptors.

Clinical features

1-Breast lump

- Commonest presenting symptom.
- Usually painless (unless inflammatory carcinoma).
- Hard and gritty feeling.
- May be immobile (held within breast tissue), tethered (attached to surrounding breast tissue or skin), or fixed (attached to chest wall).
- Ill-defined; irregular with poorly defined edges.



Clinical features

2-Nipple abnormalities

- Nipple may be the prime site of disease (Bowen's disease), presenting as an eczema-like change.
- Nipples may be affected by an underlying cancer:

-Destroyed.

-Inverted.

-Deviated.

-Associated bloody discharge.



Blood-stained nipple discharge.

Clinical features

3-Skin changes

- Carcinoma beneath skin causes dimpling, puckering, or colour changes.
- Late presentation may be with skin ulceration or fungation of the carcinoma through the skin.
- Lymphoedema of the skin (*peau d'orange*) suggests local lymph node involvement *or locally advanced cancer*.
- Extensive inflammatory changes of the skin are associated with inflammatory carcinoma (aggressive form).



Skin dimpling in the lower inner quadrant of the left breast associated with breast cancer.



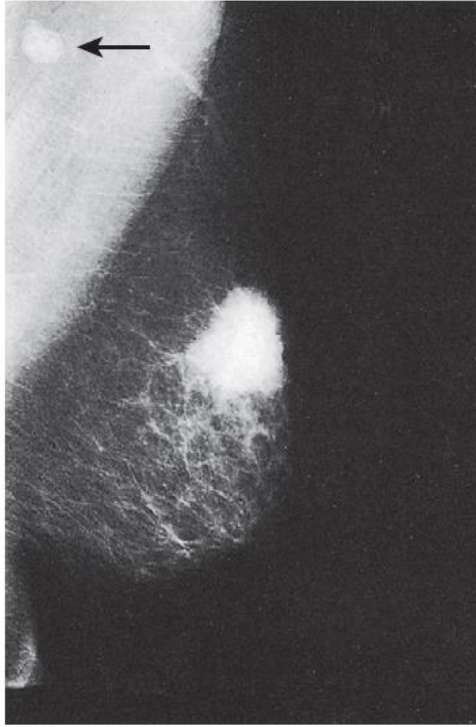
Asymmetric nipple inversion in a patient with breast cancer.

Clinical features

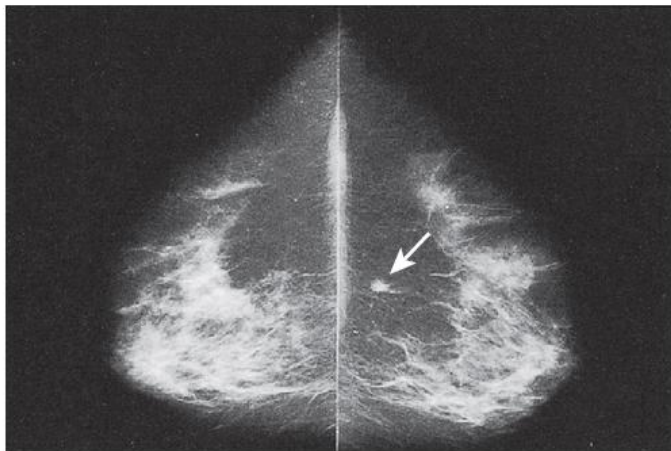
4-Systemic features

- Systemic features include weight loss, anorexia, bone pain, jaundice, malignant pleural, pericardial effusions, and anaemia.

Diagnosis & investigations

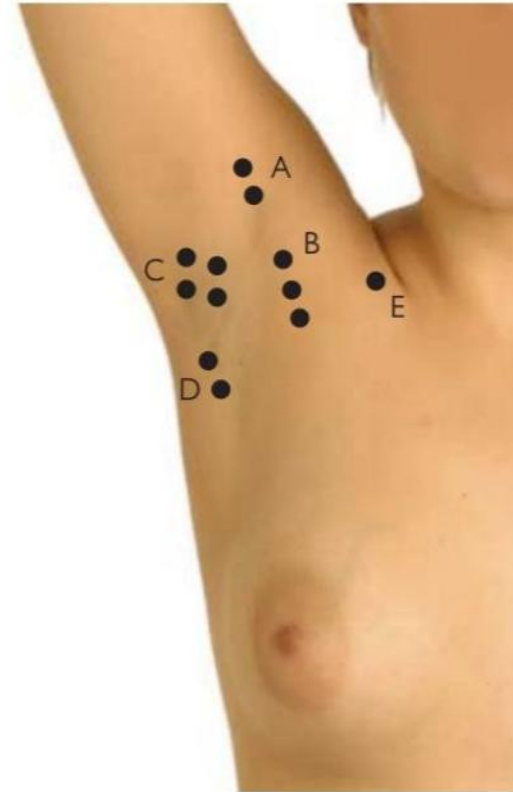
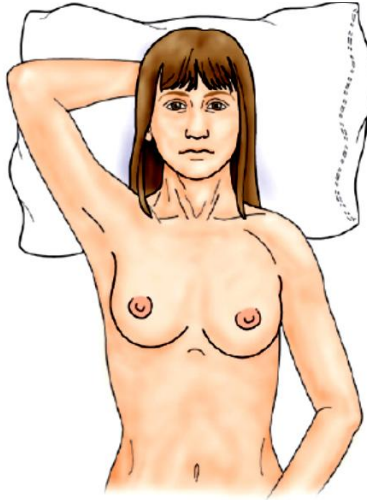
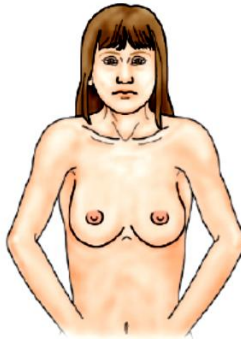
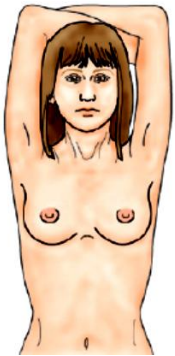
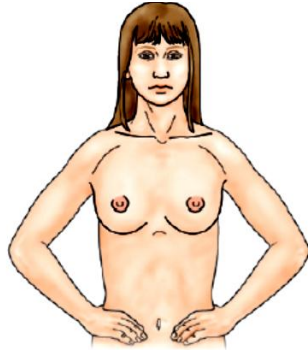
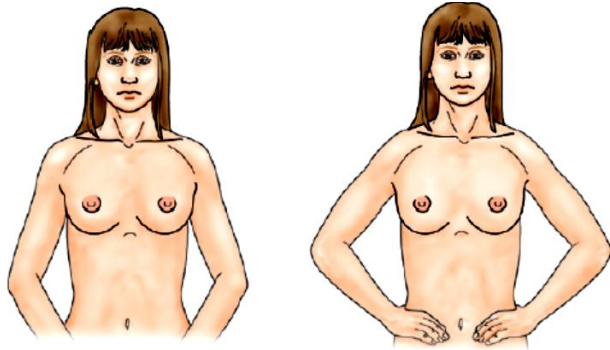


↓ Mammogram of a cancer (irregular dense mass) ar



... Mammogram of a cancer detected at breast screening.
A small lesion at the back of the left breast (arrow).

- All breast lumps or suspected carcinomas are investigated with triple assessment.
- Clinical examination (as above).
- Radiological assessment:
 - *Mammography* usual, particularly over age 35y.
 - *Ultrasound* scan used to assess the presence of involved lymph nodes; sometimes used under age 35 because increased tissue density reduces sensitivity and specificity of mammography.
 - *MRI* used in lobular carcinoma to assess the extent of the disease, multifocality, and the opposite breast.
- *Younger women with dense breast tissue.* For screening purpose in patients with strong family history.



A = Lateral
B = Pectoral
C = Central
D = Subscapular
E = Infraclavicular



Investigations



Core biopsy being performed.

Tissue diagnosis

- Core biopsy or fine needle aspiration cytology (**FNAC**) of the breast lesion and axillary nodes.
- Core biopsy also finds oestrogen receptor status, differentiates between invasive carcinomas and *in situ* carcinoma (ductal carcinoma *in situ*, DCIS).

Staging investigations

Systemic staging is usually reserved for patients following surgical treatment with a tumour who are at risk of systemic disease.

- Staging CT scan (chest, abdomen, and pelvis).
- Liver ultrasound.
- Chest X-ray.
- Bone scan.
- LFTs, serum calcium.
- Specific investigations for organ-specific suspected metastases.

Surgical treatment of breast cancer

- Surgery is the mainstay of non-metastatic disease.
- Options for treatment of the primary tumour are as follows:

1-Wide local excision

- To ensure clear margins.
- Commonest procedure.
- Breast-conserving, provided the breast is adequate size and tumour location appropriate (not central/retro-areolar).
- Usually combined with local radiotherapy to residual breast to reduce risk of local recurrence.

Options for treatment of the primary tumour

2-Simple mastectomy

- Best local treatment and cosmetic result for large tumours (especially in small breast), central location, late presentation with complications such as ulceration.
- Also used for multifocal tumours or where there is evidence of widespread *in situ* changes.
- *Adjuvant breast radiotherapy is very rarely necessary.*
- **Performed with reconstruction at the same time or later stage including:**
 - Latissimus dorsi flap;
 - TRAM flap;
 - Prosthesis

—— Surgical management of regional lymph node

1-Axillary node sampling

- Minimum of four nodes should be retrieved.
- Avoids complete disruption to axillary lymph drainage, reducing risk of lymphoedema.
- Is inadequate for treatment of the axilla. If nodes are +ve, they require adjuvant radiotherapy to axilla or *axillary node clearance*.

2-Axillary node clearance

- Optimizes diagnosis and treatment of axilla.
- Increases risk of lymphoedema greatly.

3-Sentinel node biopsy

- One or two nodes primarily draining tumour identified by radioactive tracer or dye injected around tumour and node(s).
- Identify positive nodes, then require a full axillary clearance.
- Avoids major axillary surgery where not necessary.



Surgery for metastatic disease

- Surgery in metastatic disease is limited to procedures for symptomatic
- control of local disease (e.g. mastectomy to remove fungating tumour).

Ductal carcinoma *in situ* (DCIS)

- Precancerous condition.
- Ten to fifty per cent develop invasive ductal cancers.
- Mammograms show microcalcification.
- Pathologically graded to low grade, intermediate grade, and high grade.
- DCIS is treated with wide local excision with clear margin.
- Mastectomy needed in larger breast lesions or multifocal disease.
- High grade DCIS treated by post-operative radiotherapy after wide local excision.
- Axillary surgery is not needed as there is no potential for lymph node metastasis.

Medical treatment

In non-metastatic disease, medical therapy is adjuvant to reduce the risk of systemic relapse, usually after primary surgery.

It is occasionally used as a treatment of choice of elderly or those unfit/inappropriate for surgery.

1-Endocrine therapy.

- Used in (o)estrogen receptor (ER) +ve patients.
- Anti-oestrogens like tamoxifen or aromatase inhibitors (letrozole).
- Post-menopausal patients—letrozole (caution osteoporosis).
- Premenopausal patients—tamoxifen.
- Herceptin—given in Her-2 receptor +ve patients.



2-Chemotherapy (e.g. anthracyclines, cyclophosphamide, 5-FU, methotrexate). Offered to patients with high risk features (+ve nodes, poor grade, young patients).

In metastatic disease, medical therapy is palliative to increase survival time and includes:

- *Endocrine therapy.*
- *Chemotherapy* (e.g. anthracyclines, taxanes, herceptin).
- *Radiotherapy.* To reduce pain of bony metastases or symptoms from cerebral or liver disease.



Thank You